Akif Zahin

Email | Linkedin | Github | Website

Education

North South University — Dhaka, Bangladesh
*Bachelor of Science in Computer Science and Engineering — **CGPA: 3.45*** — ***Oct 2021 – Dec 2025***

South Breeze School Uttara — Dhaka, Bangladesh
O-Levels & A-Levels — ***June 2019 – June 2021***

- **O-Level Grades:** four 8s and four 9s
- **A-Level Grades:** two A*s and two As

Experience

Guardian Life Insurance Ltd | *Intern, Information Technology Department* — **May 2026 – Aug 2026**

- Rebuilt internal phonebook system using **Next.js**, **TypeScript**, and **Tailwind CSS**, improving usability and maintainability
- Revamped company website frontend with **React**, enhancing UI consistency and user experience
- Collaborated with cross-functional teams to architect and implement **MVC** design patterns for upcoming internal tools
- Designed and managed relational database schemas using **MySQL** to support new application features

Projects

NewLens | *Python, FastAPI, PyTorch, HTML, JavaScript, Docker* — **July 2024 – Dec 2024**

- Engineered a full-stack web app for real-time sentiment and political bias detection in news articles
- Fine-tuned a **BERT-based transformer model** via HuggingFace to classify bias as **Left, Right, or Neutral**
- Implemented **FastAPI REST endpoints** for model serving with asynchronous I/O for low latency
- Containerized the entire system using **Docker** for reproducible deployment
- Part of junior design project (**CSE299**)

LM-FCIL | *Python, Kotlin, TensorFlow, TFLite, Federated Learning, OpenCV* — **Jan 2025 – Aug 2025**

- Developed an Android app for **offline handwriting segmentation and recognition** using deep CNNs and OpenCV
- Implemented a **Federated Learning** framework for on-device model updates without sharing raw data
- Used **MobileNetV3 Small** as backbone; designed client-server sync protocol for decentralized training
- Part of senior research project (**CSE498R**)

TimeGaitNet | *Python, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas* — **June 2025 – Dec 2025**

- Designed a deep learning pipeline for **Parkinson's Freezing of Gait (FoG) detection** using wearable sensor data
- Leveraged **multi-task learning** and multimodal fusion of temporal and spatial gait features
- Applied extensive data augmentation to mitigate class imbalance; first novel use of state space models for FoG
- Part of senior capstone project (**CSE499B**)

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS, PHP
Frameworks: React, Next.js, Node.js, FastAPI, TensorFlow, PyTorch, OpenCV, Docker
Developer Tools: Git, VS Code, Android Studio
Libraries: pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, HuggingFace

Awards

Full Merit Based Entrance Scholarship — North South University, 2021
Daily Star Award — O-Level Academic Excellence, 2019